INCH-POUND

MIL-DTL-24308/24H
w/AMENDMENT 2
13 January 2012
SUPERSEDING
MIL-DTL-24308/24H
w/AMENDMENT 1
9 February 2011

DETAIL SPECIFICATION SHEET

# CONNECTORS, ELECTRIC, RECTANGULAR, NONENVIRONMENTAL, MINIATURE, POLARIZED SHELL, RACK AND PANEL, PIN CONTACTS PRINTED WIRING BOARD TERMINATION TYPES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24308.

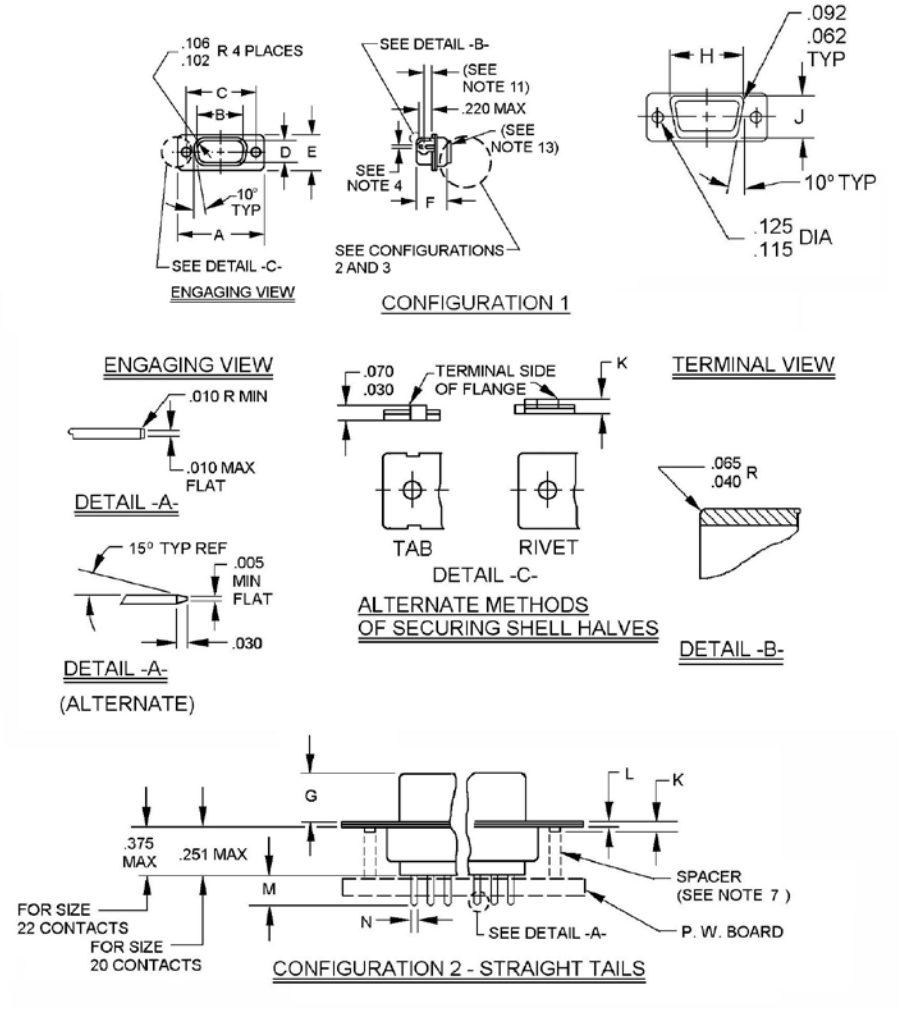



FIGURE 1. Shell, plug, printed wiring board type.



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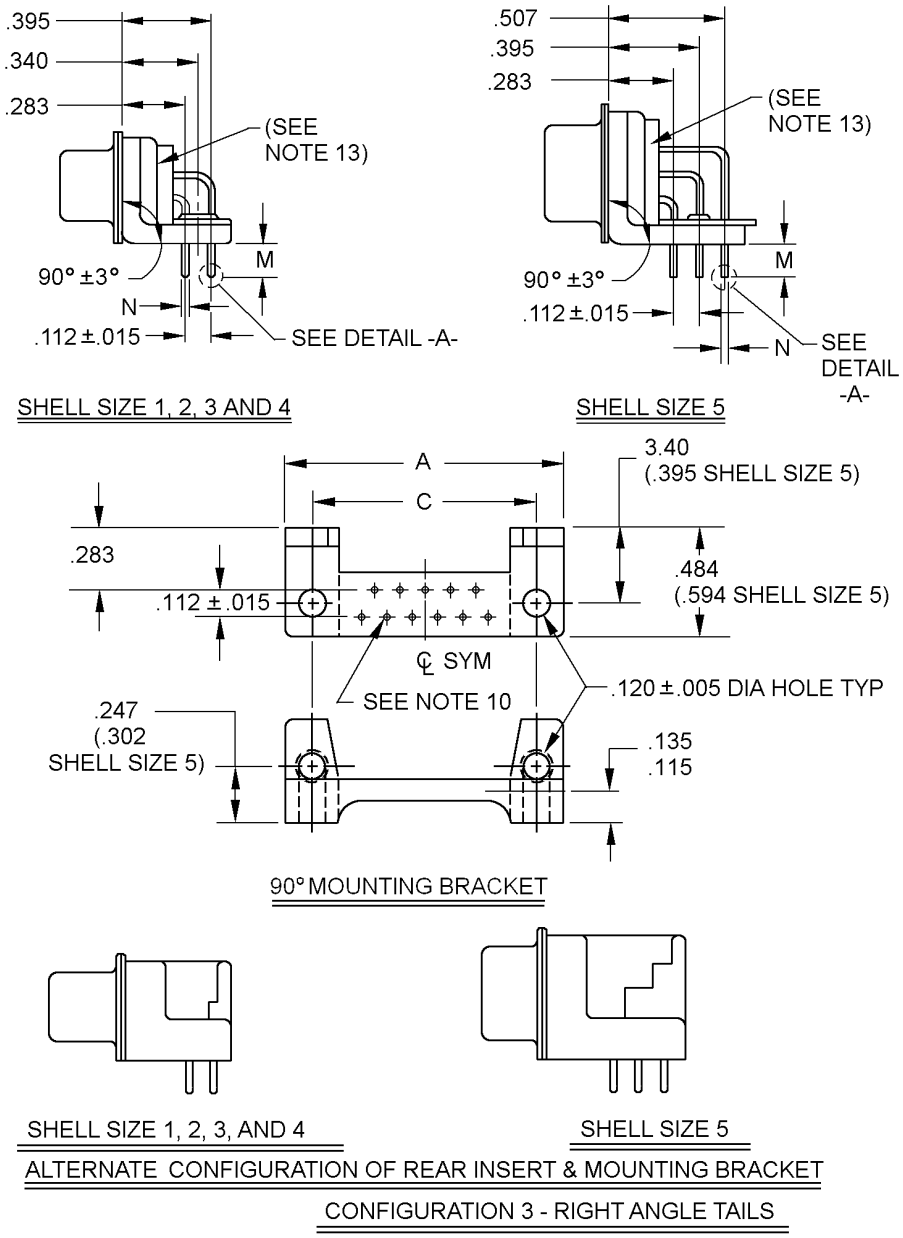



FIGURE 1. Shell, plug, printed wiring board type – Continued.



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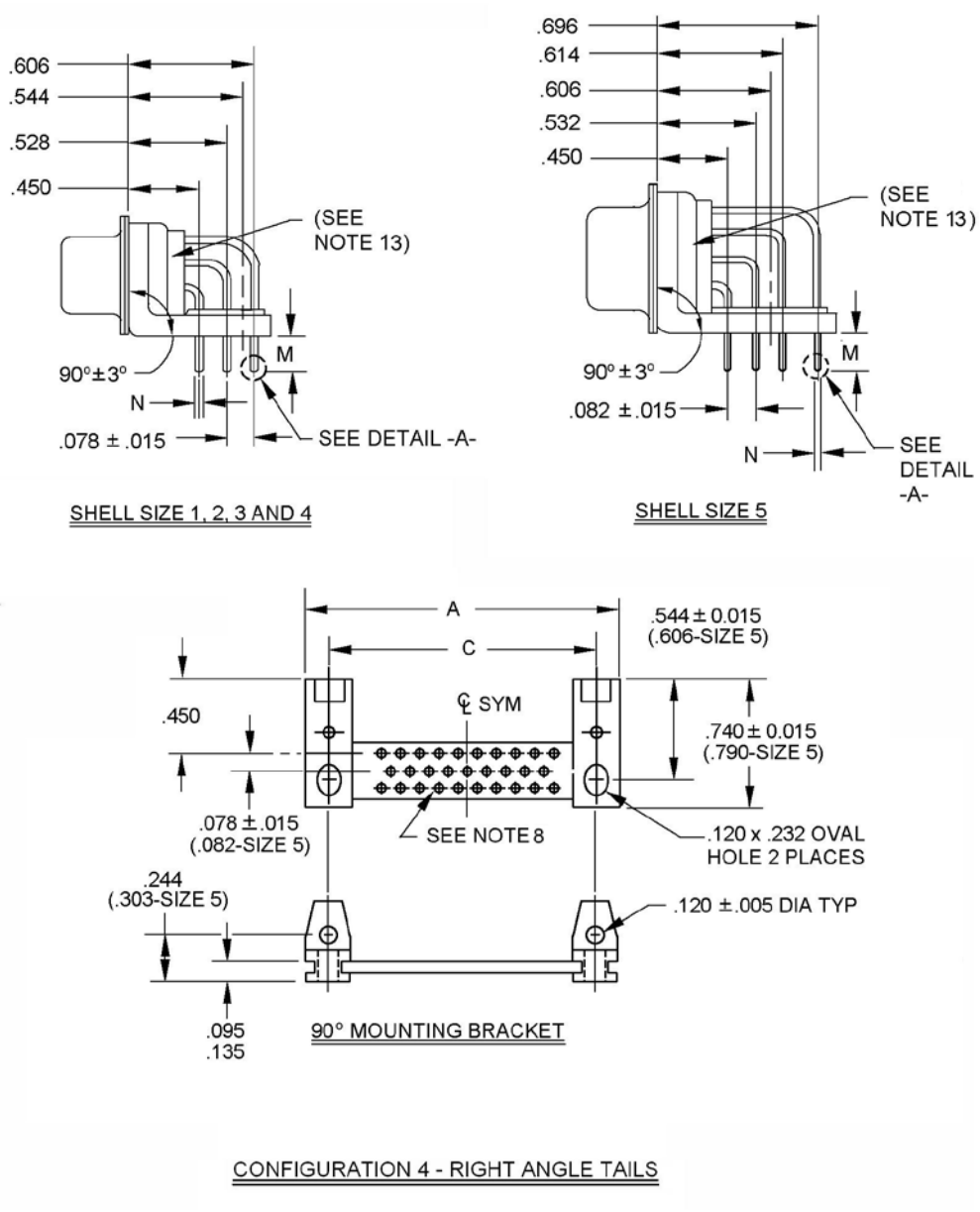



FIGURE 1. Shell, plug, printed wiring board type – Continued.



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Dimensions and configurations

| Shell Size | A | B | C | D | E | F | G | H | J | K | L |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1.228 (31.19) 1.198 (30.43) | .671 (17.04) .661 (16.79) | .989 (25.12) .979 (24.87) | .334 (8.48) .324 (8.23) | .509 (12.93) .479 (12.17) | .432 (10.97) .412 (10.46) | .238 (6.05) .229 (5.82) | .769 (19.53) .749 (19.02) | .432 (10.97) .412 (10.46) | .060 (1.52) .035 (0.89) | .040 (1.02) .020 (0.51) |
| 2 | 1.556 (39.52) 1.526 (38.76) | .999 (25.37) .989 (25.12) | 1.317 (33.45) 1.307 (33.20) | .334 (8.48) .324 (8.23) | .509 (12.93) .479 (12.17) | .432 (10.97) .412 (10.46) | .238 (6.05) .229 (5.82) | 1.093 (27.76) 1.073 (27.25) | .432 (10.97) .412 (10.46) | .060 (1.52) .035 (0.89) | .040 (1.02) .020 (0.51) |
| 3 | 2.103 (53.42) 2.073 (52.65) | 1.539 (39.09) 1.529 (38.84) | 1.857 (47.17) 1.847 (46.91) | .334 (8.48) .324 (8.23) | .509 (12.93) .479 (12.17) | .436 (11.07) .416 (10.57) | .236 (5.99) .224 (5.69) | 1.635 (41.53) 1.615) (41.02) | .432 (10.97) .412 (10.46) | .070 (1.78) .050 (1.27) | .049 (1.24) .029 (0.74) |
| 4 | 2.744 (69.70) 2.714 (68.94) | 2.187 (55.55) 2.177 (55.30) | 2.505 (63.63) 2.495 (63.37) | .334 (8.48) .324 (8.23) | .509 (12.93) .479 (12.17) | .436 (11.07) .416 (10.57) | .236 (5.99) .224 (5.69) | 2.282 (57.96) 2.262 (57.45) | .432 (10.97) .412 (10.46) | .070 (1.78) .050 (1.27) | .049 (1.24) .029 (0.74) |
| 5 | 2.650 (67.31) 2.620 (66.55) | 2.084 (52.93) 2.074 (52.68) | 2.411 (61.24) 2.401 (60.99) | .446 (11.33) .436 (11.07) | .620 (15.75) .590 (14.00) | .436 (11.07) .416 (10.57) | .236 (5.99) .224 (5.69) | 2.188 (55.58) 2.168 (55.07) | .544 (13.82) .524 (13.31) | .070 (1.78) .050 (1.27) | .049 (1.24) .029 (0.74) |

Application data and dimensions.

| Applicable dash number M24308/24 | Applicable Configurations | Applicable board thickness | Nominal pin diameter | M | N |
|---|---|---|---|---|---|
| -1 through –5 (See note 15) | 1 and 2 | .094 (2.39 mm) | .030 (0.76 mm) | .185 (4.70 mm) .130 (3.30 mm) | .033 (0.84 mm) .027 (.027 mm) |
| -7 through –11 (See note 15) | 1 and 2 | .125 (3.18 mm) | .030 (0.76 mm) | .210 (5.33 mm) .155 (3.94 mm) | .033 (0.84 mm) .027 (.027 mm) |
| -13 through –17 (See note 15) | 1 and 2 | .094 (2.39 mm) | .040 (1.02 mm) | .185 (4.70 mm) .130 (3.30 mm) | .043 (1.09 mm) .037 (0.94 mm) |
| -19 through –23 (See note 15) | 1 and 2 | .125 (3.18 mm) | .040 (1.02 mm) | .210 (5.33 mm) .155 (3.94 mm) | .043 (1.09 mm) .037 (0.94 mm) |
| -25 through –29 (See note 15)/ | 1 and 3 | .062 (1.57 mm) | .030 (0.76 mm) | .154 (3.91 mm) .099 (2.51 mm) | .033 (0.84 mm) .027 (.027 mm) |
| -31 through –35 (See note 15) | 1 and 3 | .094 (2.39 mm) | .030 (0.76 mm) | .185 (4.70 mm) .130 (3.30 mm) | .033 (0.84 mm) .027 (.027 mm) |
| -37 through –41 (See note 15) | 1 and 3 | .062 (1.57 mm) | .040 (1.02 mm) | .154 (3.91 mm) .099 (2.51 mm) | .043 (1.09 mm) .037 (0.94 mm) |
| -43 through –47 (See note 15) | 1 and 3 | .094 (2.39 mm) | .040 (1.02 mm) | .185 (4.70 mm) .130 (3.30 mm) | .043 (1.09 mm) .037 (0.94 mm) |
| -49 through –53 (See note 15) | 1 and 3 | .125 (3.18 mm) | .030 (0.76 mm) | .210 (5.33 mm) .155 (3.94 mm) | .033 (0.84 mm) .027 (.027 mm) |
| -55 through –59 (See note 15) | 1 and 3 | .125 (3.18 mm) | .040 (1.02 mm) | .210 (5.33 mm) .155 (3.94 mm) | .043 (1.09 mm) .037 (0.94 mm) |
| -61 through –65 (See note 15) | 1 and 2 | .094 (2.39 mm) | .020 (0.51 mm) | .185 (4.70 mm) .130 (3.30 mm) | .023 (0.58 mm) .017 (0.43 mm) |
| -67 through –71 (See note 15) | 1 and 2 | .125 (3.18 mm) | .020 (0.51 mm) | .320 (8.13 mm) .275 (6.99 mm) | .023 (0.58 mm) .017 (0.43 mm) |
| -73 through –77 (See note 15) | 1 and 4 | .062 (1.57 mm) | .020 (0.51 mm) | .154 (3.91 mm) .099 (2.51 mm) | .023 (0.58 mm) .017 (0.43 mm) |

FIGURE 1. Shell, plug, printed wiring board type – Continued.



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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .005 | 0.13 | .104 | 2.64 | .247 | 6.27 |
| .010 | 0.25 | 106 | 2.69 | .251 | 6.38 |
| .015 | 0.38 | .112 | 2.84 | .283 | 7.19 |
| .030 | 0.76 | .115 | 2.92 | .302 | 7.67 |
| .040 | 1.02 | .120 | 3.05 | .340 | 8.64 |
| .062 | 1.57 | .125 | 3.18 | .395 | 10.03 |
| .065 | 1.65 | .135 | 3.43 | .484 | 12.29 |
| .070 | 1.78 | .220 | 5.59 | .507 | 12.88 |
| .092 | 2.34 | | | .594 | 15.09 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Pin diameter .041/.039 (1.04/0.99 mm) diameter pin for arrangements (see MIL-DTL-24308, appendix A) A-1-1, A-2-1, A-3-1, A-4-1 and A-5-1. Pin diameter .0305/.0295 (.775/.749 mm) diameter for arrangements A-1-2, A-2-2, A-3-2, A-4-2, and A-5-2.
4. Copyright notice. All information disclosed in this military specification sheet which is or may be copyrighted by ITT Cannon Electric is reproduced herein with the expressed permission of the copyright owner.
5. The 90 ° bracket shall be one piece flame resistant glass-filled nylon, polyester or two piece metal and nylon of which the metal pieces shall be coated with a material in accordance with IPC-CC-830.
6. The 90 ° bracket shall be furnished as an integral part of the connector assembly.
7. Spacers will be supplied with connectors (not attached to the connector).
8. Layout shown for reference only to locate $C_L$. See applicable insert arrangement. Connectors may be fastened to board for test.
9. Dimensions are .150 (3.81 mm) minimum full pin diameter extension (sizes 3 through 6) and .159 (4.04 mm) minimum full pin diameter extension (sizes 1 and 2).
10. Unless otherwise specified, tolerance is ± .010 inch (0.25 mm).
11. In the event of a conflict between the text of this specification sheet and the references cited herein, the text of this specification sheet shall take precedence.
12. Referenced Governments documents of the issue listed in ASSIST Online database at https://assist.daps.dla.mil/ specified in the solicitation form a part of this standard to the extent specified herein.
13. Indentation on the ends of the termination side of the shell, for the purpose of securing the insulator in the shell, is permitted. This may result in the “F” dimension being undersized in these areas. However the “F” dimension shall be maintained for the remainder of the connector.
14. For classes M, N and D use table I for applicable dash numbers.

FIGURE 1. Shell, plug, printed wiring board type – Continued.



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REQUIREMENTS:

The connectors shall conform to figure 1 and the specification Part or Identifying Numbers (PIN) indicated in table I.

TABLE I. Design standards. 1/

| PIN Class G | PIN Class M | PIN Class N | PIN Class D | Shell size | Shell detail configurations | Insert arrangement reference number 2/ | Contact | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Mating end | PWB tail |
| M24308/24-1 | M24308/24-101 | M24308/24-201 | M24308/24-301 | 1 | 1 and 2 | A-1-1 | Size 20 | Straight |
| M24308/24-2 | M24308/24-102 | M24308/24-202 | M24308/24-302 | 2 | 1 and 2 | A-2-1 | Size 20 | Straight |
| M24308/24-3 | M24308/24-103 | M24308/24-203 | M24308/24-303 | 3 | 1 and 2 | A-3-1 | Size 20 | Straight |
| M24308/24-4 | M24308/24-104 | M24308/24-204 | M24308/24-304 | 4 | 1 and 2 | A-4-1 | Size 20 | Straight |
| M24308/24-5 | M24308/24-105 | M24308/24-205 | M24308/24-305 | 5 | 1 and 2 | A-5-1 | Size 20 | Straight |
| M24308/24-7 | M24308/24-107 | M24308/24-207 | M24308/24-307 | 1 | 1 and 2 | A-1-1 | Size 20 | Straight |
| M24308/24-8 | M24308/24-108 | M24308/24-208 | M24308/24-308 | 2 | 1 and 2 | A-2-1 | Size 20 | Straight |
| M24308/24-9 | M24308/24-109 | M24308/24-209 | M24308/24-309 | 3 | 1 and 2 | A-3-1 | Size 20 | Straight |
| M24308/24-10 | M24308/24-110 | M24308/24-210 | M24308/24-310 | 4 | 1 and 2 | A-4-1 | Size 20 | Straight |
| M24308/24-11 | M24308/24-111 | M24308/24-211 | M24308/24-311 | 5 | 1 and 2 | A-5-1 | Size 20 | Straight |
| M24308/24-13 | M24308/24-113 | M24308/24-213 | M24308/24-313 | 1 | 1 and 2 | A-1-1 | Size 20 | Straight |
| M24308/24-14 | M24308/24-114 | M24308/24-214 | M24308/24-314 | 2 | 1 and 2 | A-2-1 | Size 20 | Straight |
| M24308/24-15 | M24308/24-115 | M24308/24-215 | M24308/24-315 | 3 | 1 and 2 | A-3-1 | Size 20 | Straight |
| M24308/24-16 | M24308/24-116 | M24308/24-216 | M24308/24-316 | 4 | 1 and 2 | A-4-1 | Size 20 | Straight |
| M24308/24-17 | M24308/24-117 | M24308/24-217 | M24308/24-317 | 5 | 1 and 2 | A-5-1 | Size 20 | Straight |
| M24308/24-19 | M24308/24-119 | M24308/24-219 | M24308/24-319 | 1 | 1 and 2 | A-1-1 | Size 20 | Straight |
| M24308/24-20 | M24308/24-120 | M24308/24-220 | M24308/24-320 | 2 | 1 and 2 | A-2-1 | Size 20 | Straight |
| M24308/24-21 | M24308/24-121 | M24308/24-221 | M24308/24-321 | 3 | 1 and 2 | A-3-1 | Size 20 | Straight |
| M24308/24-22 | M24308/24-122 | M24308/24-222 | M24308/24-322 | 4 | 1 and 2 | A-4-1 | Size 20 | Straight |
| M24308/24-23 | M24308/24-123 | M24308/24-223 | M24308/24-323 | 5 | 1 and 2 | A-5-1 | Size 20 | Straight |
| M24308/24-25 | M24308/24-125 | M24308/24-225 | M24308/24-325 | 1 | 1 and 3 | A-1-1 | Size 20 | Rt Angle |
| M24308/24-26 | M24308/24-126 | M24308/24-226 | M24308/24-326 | 2 | 1 and 3 | A-2-1 | Size 20 | Rt Angle |
| M24308/24-27 | M24308/24-127 | M24308/24-227 | M24308/24-327 | 3 | 1 and 3 | A-3-1 | Size 20 | Rt Angle |
| M24308/24-28 | M24308/24-128 | M24308/24-228 | M24308/24-328 | 4 | 1 and 3 | A-4-1 | Size 20 | Rt Angle |
| M24308/24-29 | M24308/24-129 | M24308/24-229 | M24308/24-329 | 5 | 1 and 3 | A-5-1 | Size 20 | Rt Angle |
| M24308/24-31 | M24308/24-131 | M24308/24-231 | M24308/24-331 | 1 | 1 and 3 | A-1-1 | Size 20 | Rt Angle |
| M24308/24-32 | M24308/24-132 | M24308/24-232 | M24308/24-332 | 2 | 1 and 3 | A-2-1 | Size 20 | Rt Angle |
| M24308/24-33 | M24308/24-133 | M24308/24-233 | M24308/24-333 | 3 | 1 and 3 | A-3-1 | Size 20 | Rt Angle |
| M24308/24-34 | M24308/24-134 | M24308/24-234 | M24308/24-334 | 4 | 1 and 3 | A-4-1 | Size 20 | Rt Angle |
| M24308/24-35 | M24308/24-135 | M24308/24-235 | M24308/24-335 | 5 | 1 and 3 | A-5-1 | Size 20 | Rt Angle |
| M24308/24-37 | M24308/24-137 | M24308/24-237 | M24308/24-337 | 1 | 1 and 3 | A-1-1 | Size 20 | Rt Angle |
| M24308/24-38 | M24308/24-138 | M24308/24-238 | M24308/24-338 | 2 | 1 and 3 | A-2-1 | Size 20 | Rt Angle |
| M24308/24-39 | M24308/24-139 | M24308/24-239 | M24308/24-339 | 3 | 1 and 3 | A-3-1 | Size 20 | Rt Angle |
| M24308/24-40 | M24308/24-140 | M24308/24-240 | M24308/24-340 | 4 | 1 and 3 | A-4-1 | Size 20 | Rt Angle |
| M24308/24-41 | M24308/24-141 | M24308/24-241 | M24308/24-341 | 5 | 1 and 3 | A-5-1 | Size 20 | Rt Angle |
| M24308/24-43 | M24308/24-143 | M24308/24-243 | M24308/24-343 | 1 | 1 and 3 | A-1-1 | Size 20 | Rt Angle |
| M24308/24-44 | M24308/24-144 | M24308/24-244 | M24308/24-344 | 2 | 1 and 3 | A-2-1 | Size 20 | Rt Angle |
| M24308/24-45 | M24308/24-145 | M24308/24-245 | M24308/24-345 | 3 | 1 and 3 | A-3-1 | Size 20 | Rt Angle |
| M24308/24-46 | M24308/24-146 | M24308/24-246 | M24308/24-346 | 4 | 1 and 3 | A-4-1 | Size 20 | Rt Angle |
| M24308/24-47 | M24308/24-147 | M24308/24-247 | M24308/24-347 | 5 | 1 and 3 | A-5-1 | Size 20 | Rt Angle |

See footnotes at end of table.



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TABLE I. Design standards- Continued. 1/

| PIN Class G | PIN Class M | PIN Class N | PIN Class D | Shell size | Shell detail configurations | Insert arrangement reference number 2/ | Contact | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Mating end | PWB tail |
| M24308/24-49 | M24308/24-149 | M24308/24-249 | M24308/24-349 | 1 | 1 and 3 | A-1-1 | Size 20 | Rt Angle |
| M24308/24-50 | M24308/24-150 | M24308/24-250 | M24308/24-350 | 2 | 1 and 3 | A-2-1 | Size 20 | Rt Angle |
| M24308/24-51 | M24308/24-151 | M24308/24-251 | M24308/24-351 | 3 | 1 and 3 | A-3-1 | Size 20 | Rt Angle |
| M24308/24-52 | M24308/24-152 | M24308/24-252 | M24308/24-352 | 4 | 1 and 3 | A-4-1 | Size 20 | Rt Angle |
| M24308/24-53 | M24308/24-153 | M24308/24-253 | M24308/24-353 | 5 | 1 and 3 | A-5-1 | Size 20 | Rt Angle |
| M24308/24-55 | M24308/24-155 | M24308/24-255 | M24308/24-355 | 1 | 1 and 3 | A-1-1 | Size 20 | Rt Angle |
| M24308/24-56 | M24308/24-156 | M24308/24-256 | M24308/24-356 | 2 | 1 and 3 | A-2-1 | Size 20 | Rt Angle |
| M24308/24-57 | M24308/24-157 | M24308/24-257 | M24308/24-357 | 3 | 1 and 3 | A-3-1 | Size 20 | Rt Angle |
| M24308/24-58 | M24308/24-158 | M24308/24-258 | M24308/24-358 | 4 | 1 and 3 | A-4-1 | Size 20 | Rt Angle |
| M24308/24-59 | M24308/24-159 | M24308/24-259 | M24308/24-359 | 5 | 1 and 3 | A-5-1 | Size 20 | Rt Angle |
| M24308/24-61 | M24308/24-161 | M24308/24-261 | M24308/24-361 | 1 | 1 and 2 | A-1-2 | Size 22 | Straight |
| M24308/24-62 | M24308/24-162 | M24308/24-262 | M24308/24-362 | 2 | 1 and 2 | A-2-2 | Size 22 | Straight |
| M24308/24-63 | M24308/24-163 | M24308/24-263 | M24308/24-363 | 3 | 1 and 2 | A-3-2 | Size 22 | Straight |
| M24308/24-64 | M24308/24-164 | M24308/24-264 | M24308/24-364 | 4 | 1 and 2 | A-4-2 | Size 22 | Straight |
| M24308/24-65 | M24308/24-165 | M24308/24-265 | M24308/24-365 | 5 | 1 and 2 | A-5-2 | Size 22 | Straight |
| M24308/24-67 | M24308/24-167 | M24308/24-267 | M24308/24-367 | 1 | 1 and 2 | A-1-2 | Size 22 | Straight |
| M24308/24-68 | M24308/24-168 | M24308/24-268 | M24308/24-368 | 2 | 1 and 2 | A-2-2 | Size 22 | Straight |
| M24308/24-69 | M24308/24-169 | M24308/24-269 | M24308/24-369 | 3 | 1 and 2 | A-3-2 | Size 22 | Straight |
| M24308/24-70 | M24308/24-170 | M24308/24-270 | M24308/24-370 | 4 | 1 and 2 | A-4-2 | Size 22 | Straight |
| M24308/24-71 | M24308/24-171 | M24308/24-271 | M24308/24-371 | 5 | 1 and 2 | A-5-2 | Size 22 | Straight |
| M24308/24-73 | M24308/24-173 | M24308/24-273 | M24308/24-373 | 1 | 1 and 4 | A-1-2 | Size 22 | Rt Angle |
| M24308/24-74 | M24308/24-174 | M24308/24-274 | M24308/24-374 | 2 | 1 and 4 | A-2-2 | Size 22 | Rt Angle |
| M24308/24-75 | M24308/24-175 | M24308/24-275 | M24308/24-375 | 3 | 1 and 4 | A-3-2 | Size 22 | Rt Angle |
| M24308/24-76 | M24308/24-176 | M24308/24-276 | M24308/24-376 | 4 | 1 and 4 | A-4-2 | Size 22 | Rt Angle |
| M24308/24-77 | M24308/24-177 | M24308/24-277 | M24308/24-377 | 5 | 1 and 4 | A-5-2 | Size 22 | Rt Angle |

1/ These connectors mate with connectors specified in MIL-DTL-24308/1, MIL-DTL-24308/2, MIL-DTL-24308/5, and MIL-DTL-24308/23.
2/ For insert arrangements see MIL-DTL-24308, Appendix A.

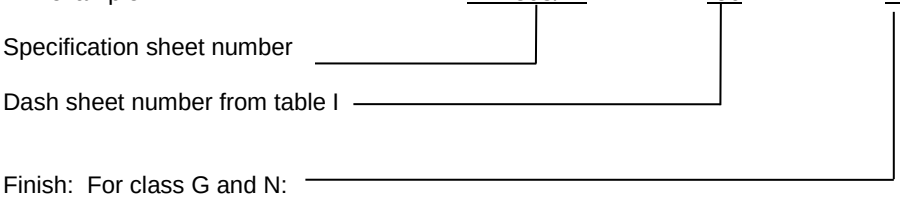
NOTE: Qualification by similarity: Qualification of connectors of one shell size of one configuration with a specific printed wiring tail length and diameter shall qualify all connectors of the same shell size and configuration with all other printed wiring tail lengths and diameters.



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PIN: Consists of the letter M, the basic number of the specifications sheet, and a dash number compiled from the code.

PIN example: M24308/24 -50 F




A = Pure electrodeposited aluminum.
F = Cadmium. ((Navy-SH prohibits the use of cadmium finished connectors for shipboard application)
K = Zinc nickel.
P = Passivated stainless steel
T = Nickel Fluorocarbon Polymer.
Z = zinc.

For class M:
Class M connectors have a gold finish, no suffix is required.

For class D:
Class D connectors have a nickel finish, no suffix is required.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-24308, this document references the following:

MIL-DTL-24308/1
MIL-DTL-24308/2
MIL-DTL-24308/5
MIL-DTL-24308/23
IPC-CC-830



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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA – NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2011-102)

Review activities:
Army - AT, AV, CR4, MI
Navy - AS, CG, MC, SH
Air Force – 99

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